

Panel Data Econometrics

Lecture 1: Foundations

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Roadmap

1. Why Panel Data?
2. The Unobserved Effects Model
3. Why Pooled OLS Fails
4. Fixed Effects as Within-Unit Identification
5. First Differences Versus Fixed Effects
6. Two-Way Fixed Effects
7. Inference and Clustering
8. Practical Diagnostics
9. Lecture 1 Summary

Where panel data methods are useful

Typical empirical settings

- **Labor:** workers observed across years; wages, training, job mobility.
- **Firm finance and productivity:** firms observed over time; investment, leverage, exports.
- **Public economics:** municipalities, schools, hospitals, or tax units tracked across periods.
- **Development and political economy:** districts, villages, households, or countries followed through reforms and shocks.
- **Industrial organization:** products, stores, or markets observed repeatedly as prices and demand move.

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Common feature

In all of these settings, units differ in ways we do not fully observe, and many of those differences are persistent.

Why researchers use panel data

What goes wrong in a cross-section

- High- x units are compared with low- x units.
- But those units also differ in latent traits:
 - ability
 - managerial quality
 - local institutions
 - baseline health
- Then the regressor is mixed with omitted heterogeneity.

What panel data improve

- We observe the same unit repeatedly.
- We can compare a unit to its own past.
- Time-invariant unit heterogeneity can be removed.
- With time effects, we can also net out aggregate shocks common to all units.

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Bottom line

Panel data are valuable because they often replace a weak *across-unit* comparison with a stronger *within-unit over time* comparison.

A motivating comparison: wages and training

Cross-sectional question

Do workers with more training tend to earn higher wages?

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Workers with more training may also differ in ability, motivation, firm quality, or family background. A high-training worker and a low-training worker are not cleanly comparable.

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Panel-data question

For the *same worker*, when training rises, does wage rise relative to that worker's own baseline?

$$Wage_{it} = \beta Train_{it} + \alpha_i + u_{it}$$

If worker ability is stable over time, α_i can be removed by fixed effects or first differences.

What changes when data become a panel?

Three canonical data structures

- **Cross-section:** many units, one date.
- **Time series:** one unit, many dates.
- **Panel:** many units, many dates.

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The central econometric question

Empirical work is about comparisons. Panel data let us compare:

- one unit to another unit, and
- the same unit to itself over time.

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Fixed-effects intuition

Instead of asking whether high- x units have high y , panel data let us ask:

When a given unit changes x_{it} , does y_{it} also change?

Within and between variation

For any regressor x_{it} , define the unit mean and grand mean:

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^T x_{it}, \quad \bar{x} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T x_{it}.$$

Then every observation decomposes as

$$x_{it} = \bar{x} + (\bar{x}_i - \bar{x}) + (x_{it} - \bar{x}_i).$$

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Interpretation

- $\bar{x}$: overall level
- $\bar{x}_i - \bar{x}$: **between-unit** variation
- $x_{it} - \bar{x}_i$: **within-unit** variation

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Why this decomposition matters

Pooled OLS mixes between and within information. Fixed effects discards the between component and estimates the slope from within-unit variation only.

What panel data help with and what they do not

Main gains

1. Control for time-invariant heterogeneity.
2. Study adjustment over time.
3. Separate level differences from changes.
4. Make the source of identification transparent.

Not automatically solved

- time-varying omitted variables
- reverse causality
- anticipation effects
- measurement error
- endogenous attrition
- spillovers

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Wage example

With repeated observations on workers, we can compare a worker to her own baseline and remove persistent traits such as ability or motivation, provided those traits are roughly stable over time.

Balanced, unbalanced, and the strategic value of panels

Definitions

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 - Most estimators adapt to unbalanced panels, but missingness can itself become an econometric problem.

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Right way to think about panel data

Panel data are not a magic technique. They are a chance to ask a better comparison question: compare a unit to its own past, and when needed also net out shocks common to all units.

The workhorse model

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it}$$

Objects in the model

- $x'_{it}\beta$: observed covariates
- α_i : time-invariant unit effect
- λ_t : common time effect
- u_{it} : idiosyncratic disturbance

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Economic content

- α_i may be ability, managerial talent, soil quality, baseline health, or institutional quality.
- λ_t may be inflation, recession, legislation, or technology shocks common to all units.

Why the unit effect is central

Suppress time effects for a moment:

$$y_{it} = \beta x_{it} + \alpha_i + u_{it}.$$

Define the pooled error

$$v_{it} = \alpha_i + u_{it}.$$

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The key failure mode

Pooled OLS requires

$$\text{Cov}(x_{it}, v_{it}) = \text{Cov}(x_{it}, \alpha_i) + \text{Cov}(x_{it}, u_{it}) = 0.$$

So even if $\text{Cov}(x_{it}, u_{it}) = 0$, pooled OLS fails whenever

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Interpretation

The difficult problem is not that units differ. It is that the permanent differences across units are often correlated with the regressors we care about.

Strict exogeneity in static panels

The standard fixed-effects condition is

$$\mathbb{E}[u_{it} \mid \alpha_i, \lambda_t, x_{i1}, \dots, x_{iT}] = 0 \quad \text{for each } t.$$

What makes this strong?

It requires orthogonality to the **entire regressor path**, not only to current regressors.

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If a positive productivity shock today causes the firm to hire more workers next year, then u_{it} predicts $x_{i,t+1}$. Contemporaneous exogeneity may hold while strict exogeneity fails.

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Three questions to ask in every application

1. What economic content sits inside α_i ?
2. Is α_i plausibly correlated with the regressors?
3. After removing α_i and perhaps λ_t , what variation is left to identify β ?

What pooled OLS is doing in a panel

Ignoring the panel structure gives

$$y_{it} = x'_{it}\beta + v_{it}, \quad v_{it} = \alpha_i + \lambda_t + u_{it}.$$

Pooled OLS compares

- one unit to another unit in the same period,
- the same unit to itself over time,
- and all mixtures of those two comparisons.

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- and all mixtures of those two comparisons.

Why this is dangerous

If cross-unit comparisons are contaminated by unobserved heterogeneity, pooled OLS blends credible within-unit information with contaminated between-unit information.

Probability limit of pooled OLS

In the scalar model

$$y_{it} = \beta x_{it} + \alpha_i + u_{it},$$

the pooled OLS estimator satisfies

$$\text{plim } \hat{\beta}^{POLS} = \beta + \frac{\text{Cov}(x_{it}, \alpha_i)}{\text{Var}(x_{it})} + \frac{\text{Cov}(x_{it}, u_{it})}{\text{Var}(x_{it})}.$$

Immediate implication

Even if $\text{Cov}(x_{it}, u_{it}) = 0$, pooled OLS is inconsistent whenever $\text{Cov}(x_{it}, \alpha_i) \neq 0$.

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Wage panel

If higher-ability workers both train more and earn more, pooled OLS attributes part of the return to ability to the training regressor.

Proof

Pooled OLS mixes between and within covariance

For any balanced panel,

$$\text{Cov}(x_{it}, y_{it}) = \text{Cov}(\bar{x}_i, \bar{y}_i) + \text{Cov}(x_{it} - \bar{x}_i, y_{it} - \bar{y}_i).$$

Two components inside the pooled slope

1. **Between covariance:** high- x units versus low- x units
2. **Within covariance:** a unit versus its own average

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Two components inside the pooled slope

1. **Between covariance:** high- x units versus low- x units
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Core lesson

If the between component is contaminated by correlation between $\bar{x}_i$ and α_i , pooled OLS answers the wrong empirical question even when panel data are available.

Proof

When pooled OLS can still be informative

Possible uses

- α_i is genuinely uncorrelated with regressors.
- The goal is descriptive rather than causal.
- The panel is very short and there is almost no within variation.

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But the usual presumption in observational work

Permanent unit heterogeneity is often correlated with observed choices. In that setting, pooled OLS should be treated as a baseline comparison, not as the preferred causal estimator.

Two views of fixed effects

Start from

$$y_{it} = x'_{it}\beta + \alpha_i + u_{it}.$$

Dummy-variable view

$$y_{it} = x'_{it}\beta + \sum_{j=1}^{N-1} \delta_j 1\{i = j\} + u_{it}.$$

Each unit gets its own intercept.

Within view

Subtract each unit's time average:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)' \beta + (u_{it} - \bar{u}_i).$$

Only within-unit movements remain.

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Identification

Fixed effects ask: when the same unit is above its own usual level of x , is it also above its own usual level of y ?

The within transformation

Define demeaned variables

$$\ddot{y}_{it} = y_{it} - \bar{y}_i, \quad \ddot{x}_{it} = x_{it} - \bar{x}_i, \quad \ddot{u}_{it} = u_{it} - \bar{u}_i.$$

Then the transformed regression is

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Applying OLS to the transformed equation yields

$$\hat{\beta}^{FE} = \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it} \ddot{x}_{it}' \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it} \ddot{y}_{it} \right).$$

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Scalar case

$$\hat{\beta}^{FE} = \frac{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - \bar{x}_i)^2}.$$

What fixed effects keep and what they discard

Kept

Within-unit variation in x_{it} and y_{it} .

Discarded

- time-invariant unit heterogeneity α_i
- all purely between-unit variation
- any regressor that never changes within a unit

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Important implication

If a regressor has lots of cross-sectional variation but barely moves within units, pooled OLS may look precise while FE is noisy. That is not a software problem. It is a lack-of-identification-variation problem.

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LSDV = FE

Why FE is consistent under strict exogeneity

Under strict exogeneity,

$$\mathbb{E}[u_{it} \mid \alpha_i, x_{i1}, \dots, x_{iT}] = 0 \quad \text{for each } t,$$

the transformed regressor is orthogonal to the transformed error:

$$\mathbb{E}[\ddot{x}_{it} \ddot{u}_{it}] = 0.$$

Consequence

If the within second-moment matrix is nonsingular, the FE estimator is consistent as $N \rightarrow \infty$ with fixed T .

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What FE does not solve

- omitted variables that change within unit over time
- reverse causality
- measurement error

Orthogonality Proof

First differencing removes the same unit effect

From

$$y_{it} = x'_{it}\beta + \alpha_i + u_{it},$$

take adjacent differences:

$$\Delta y_{it} = \Delta x'_{it}\beta + \Delta u_{it}, \quad t = 2, \dots, T.$$

Interpretation

- FE compares each period to the unit's average over the sample.
- FD compares each period only to the immediately preceding period.

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Economic language

FE asks whether the unit is above or below its usual level. FD asks whether the unit rose or fell relative to last period.

When FE and FD coincide and when they do not

Exact identity when $T = 2$

With only two periods, FE and FD extract the same single within-unit movement, so they produce the same slope coefficient.

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Why they differ for $T > 2$

FE uses all periods simultaneously through the unit mean. FD uses only adjacent comparisons. The filters react differently to trends, persistence, and measurement noise.

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Why they differ for $T > 2$

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Decision rule

Neither estimator dominates universally. The right choice depends on the serial correlation of the shocks, the measurement quality of x_{it} , and the economic meaning of within-unit variation.

Proof for $T = 2$

Why differencing changes the error process

Even if u_{it} is serially uncorrelated,

$$\Delta u_{it} = u_{it} - u_{i,t-1}$$

is mechanically serially correlated:

$$\text{Var}(\Delta u_{it}) = 2\sigma_u^2, \quad \text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) = -\sigma_u^2.$$

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Measurement-error warning

If

$$x_{it}^{obs} = x_{it}^* + \nu_{it},$$

then differencing gives

$$\Delta x_{it}^{obs} = \Delta x_{it}^* + (\nu_{it} - \nu_{i,t-1}),$$

so the transformed regressor contains two measurement-error draws.

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Practical lesson

First differencing can be appealing when the economic question is about changes, but it can be fragile when the regressor is noisy.

Why one-way fixed effects are often not enough

The two-way model is

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it}.$$

Roles

- α_i : permanent differences across units
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Firm investment and cash flow

Firm fixed effects remove permanent differences in productivity or management quality. They do not remove the boom year that simultaneously raises sales, cash flow, and investment for everyone.

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Interpretation

Two-way FE asks a stricter question: how does a unit move relative to its own average after removing what was common to everyone in that period?

Double demeaning

Define

$$\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{y}_t = \frac{1}{N} \sum_{i=1}^N y_{it}, \quad \bar{y} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{it},$$

and analogously for x_{it} .

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and analogously for x_{it} .

The two-way within transformation is

$$\tilde{y}_{it} = y_{it} - \bar{y}_i - \bar{y}_t + \bar{y}, \quad \tilde{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}.$$

Why add back the grand mean?

Because subtracting both the row mean and the column mean removes the overall level twice.

Proof

What survives under two-way fixed effects?

Under two-way FE, the identifying variation is the part of x_{it} that is

1. not explained by the unit's average level, and
2. not explained by the period's average level.

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Three implications

- Time-invariant regressors disappear.
- Purely aggregate regressors disappear once time effects are included.
- Only idiosyncratic within-unit, within-time-cell variation survives both sweeps.

What two-way FE do and do not solve

Removed

- time-invariant unit heterogeneity
- common period shocks

Still left

- unit-specific time-varying confounders
- feedback to future regressors
- staggered-adoption heterogeneity issues

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Why this matters for later topics

Two-way FE is the backbone of later designs such as difference-in-differences and event studies, so students need to know exactly what it absorbs and what it does not.

Why panel inference is different

After pooling, demeaning, differencing, or adding time effects, the variance still has the generic form

$$\text{Var}(\hat{\beta} \mid X^*) = (X^{*'} X^*)^{-1} X^{*'} \Omega X^* (X^{*'} X^*)^{-1}.$$

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If errors may be arbitrarily correlated within unit, the cluster-robust variance estimator is

$$\widehat{\text{Var}}(\hat{\beta}) = (X^{*'} X^*)^{-1} \left(\sum_{i=1}^N X_i^{*'} \hat{u}_i^* \hat{u}_i^{*'} X_i^* \right) (X^{*'} X^*)^{-1}.$$

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Key conceptual difference

Clustering keeps residual cross-products within unit instead of forcing them to zero.

Why clustering is usually the default

Even if the original disturbances are serially uncorrelated, demeaning creates dependence:

$$\ddot{u}_{it} = u_{it} - \bar{u}_j.$$

Then for $t \neq s$,

$$\text{Var}(\ddot{u}_{it}) = \sigma_u^2 \left(1 - \frac{1}{T}\right), \quad \text{Cov}(\ddot{u}_{it}, \ddot{u}_{is}) = -\frac{\sigma_u^2}{T}.$$

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Implication

Fixed effects remove α_i , but they do not generally deliver independent residuals.

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Applied rule

In worker-year panels cluster by worker; in firm-quarter panels often by firm; if dependence also runs across units within a year, consider time clustering or two-way clustering.

Derivation

Few clusters and inferential design

Few-cluster warning

Cluster asymptotics rely on the number of clusters, not the number of rows. A panel with 40,000 observations but only 18 state clusters is still an 18-cluster inference problem.

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- Report which dimension is clustered.
- Use cluster-based small-sample adjustments when needed.
- In very small-cluster settings, the wild cluster bootstrap is often attractive.

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Common mistake

“Robust standard errors” is too vague in panel work. Heteroskedasticity-robust but unclustered errors often overstate precision because they act as if dependent observations were independent.

A short checklist for applied panel work

1. What variation identifies the coefficient?
2. Does the regressor move enough within units?
3. Are time effects needed to absorb common shocks?
4. Which observations may share shocks?
5. How many independent clusters are there?
6. Do the results survive reasonable FE and clustering choices?

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Two distinctions students should keep straight

- **Time effects** address omitted aggregate shocks.
- **Clustering** addresses dependent residuals and correct uncertainty quantification.

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How to read specification changes

Comparing pooled OLS, FE, FD, and FE with time effects is informative because each estimator uses different variation. Large coefficient shifts are substantive evidence about identification, not mere robustness ritual.

R companions for Lecture 1

The notes pair Lecture 1 with small scripts in `lectures/code/panel-data/`:

- `01_within_between_variation.R`
- `02_pooled_ols_vs_fixed_effects.R`
- `03_first_differences_vs_fixed_effects.R`
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Purpose

Each script isolates one idea at a time: decomposition of variation, pooled-OLS bias, FE versus FD, common time shocks, and naive versus clustered uncertainty.

Lecture 1 summary

1. Panel data matter because they let us compare units to themselves over time.
2. Pooled OLS usually fails by mixing contaminated between variation with within variation.
3. Fixed effects remove time-invariant unit heterogeneity by demeaning.
4. First differences remove the same heterogeneity by comparing adjacent periods.
5. Two-way fixed effects additionally net out common time shocks.
6. Credible panel inference usually requires clustered standard errors.

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1. Panel data matter because they let us compare units to themselves over time.
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Unifying principle

Panel econometrics is about isolating variation that remains credible after removing persistent heterogeneity, then measuring uncertainty using the number of genuinely independent pieces of information rather than the raw number of rows.

Appendix: Proof of pooled-OLS bias formula

For a scalar regressor, the population slope from regressing y_{it} on x_{it} is

$$\frac{\text{Cov}(x_{it}, y_{it})}{\text{Var}(x_{it})}.$$

Substitute $y_{it} = \beta x_{it} + \alpha_i + u_{it}$:

$$\begin{aligned}\text{Cov}(x_{it}, y_{it}) &= \text{Cov}(x_{it}, \beta x_{it} + \alpha_i + u_{it}) \\ &= \beta \text{Var}(x_{it}) + \text{Cov}(x_{it}, \alpha_i) + \text{Cov}(x_{it}, u_{it}).\end{aligned}$$

Divide by $\text{Var}(x_{it})$:

$$\text{plim } \hat{\beta}^{POLS} = \beta + \frac{\text{Cov}(x_{it}, \alpha_i)}{\text{Var}(x_{it})} + \frac{\text{Cov}(x_{it}, u_{it})}{\text{Var}(x_{it})}.$$

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Appendix: Proof of the covariance decomposition

Write

$$x_{it} = \bar{x}_i + \tilde{x}_{it}, \quad y_{it} = \bar{y}_i + \tilde{y}_{it},$$

where $\tilde{x}_{it} = x_{it} - \bar{x}_i$ and $\tilde{y}_{it} = y_{it} - \bar{y}_i$. Then

$$\begin{aligned} \text{Cov}(x_{it}, y_{it}) &= \text{Cov}(\bar{x}_i + \tilde{x}_{it}, \bar{y}_i + \tilde{y}_{it}) \\ &= \text{Cov}(\bar{x}_i, \bar{y}_i) + \text{Cov}(\bar{x}_i, \tilde{y}_{it}) + \text{Cov}(\tilde{x}_{it}, \bar{y}_i) + \text{Cov}(\tilde{x}_{it}, \tilde{y}_{it}). \end{aligned}$$

Because $\sum_{t=1}^T (y_{it} - \bar{y}_i) = 0$ for every unit, the two cross terms are zero, so

$$\text{Cov}(x_{it}, y_{it}) = \text{Cov}(\bar{x}_i, \bar{y}_i) + \text{Cov}(\tilde{x}_{it}, \tilde{y}_{it}).$$

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Appendix: Why LSDV and within estimation coincide

Consider

$$\min_{\beta, \alpha_1, \dots, \alpha_N} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - x'_{it}\beta - \alpha_i)^2.$$

For fixed β , the first-order condition for α_i gives

$$\hat{\alpha}_i(\beta) = \bar{y}_i - \bar{x}'_i\beta.$$

Substitute back into the objective:

$$\min_{\beta} \sum_{i=1}^N \sum_{t=1}^T [(y_{it} - \bar{y}_i) - (x_{it} - \bar{x}_i)' \beta]^2.$$

This is exactly the OLS criterion from regressing $\ddot{y}_{it}$ on $\ddot{x}_{it}$, so the slope coefficients coincide.

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Appendix: Orthogonality behind fixed effects

Because $\ddot{x}_{it}$ is a function of the full regressor path $(x_{i1}, \dots, x_{iT})$,

$$\mathbb{E}[\ddot{x}_{it} u_{is}] = \mathbb{E}[\ddot{x}_{it} \mathbb{E}[u_{is} \mid \alpha_i, x_{i1}, \dots, x_{iT}]] = 0$$

for every s under strict exogeneity. Therefore

$$\mathbb{E}[\ddot{x}_{it} \bar{u}_i] = \frac{1}{T} \sum_{s=1}^T \mathbb{E}[\ddot{x}_{it} u_{is}] = 0.$$

Taking $s = t$ also gives $\mathbb{E}[\ddot{x}_{it} u_{it}] = 0$. Hence

$$\mathbb{E}[\ddot{x}_{it} \ddot{u}_{it}] = \mathbb{E}[\ddot{x}_{it} (u_{it} - \bar{u}_i)] = 0.$$

This is the moment condition that underpins FE consistency.

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Appendix: FE equals FD when $T = 2$

When $T = 2$,

$$\bar{y}_i = \frac{y_{i1} + y_{i2}}{2}, \quad \bar{x}_i = \frac{x_{i1} + x_{i2}}{2}.$$

So

$$y_{i2} - \bar{y}_i = \frac{y_{i2} - y_{i1}}{2} = \frac{\Delta y_{i2}}{2}, \quad y_{i1} - \bar{y}_i = -\frac{\Delta y_{i2}}{2},$$

and similarly

$$x_{i2} - \bar{x}_i = \frac{\Delta x_{i2}}{2}, \quad x_{i1} - \bar{x}_i = -\frac{\Delta x_{i2}}{2}.$$

Every demeaned observation is just a constant multiple of the differenced observation, so numerator and denominator scale by the same factor and the FE and FD slope coefficients are identical.

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Appendix: Why differenced errors are serially correlated

If $\Delta u_{it} = u_{it} - u_{i,t-1}$ and the original errors are serially uncorrelated with variance σ_u^2 , then

$$\begin{aligned}\text{Var}(\Delta u_{it}) &= \text{Var}(u_{it}) + \text{Var}(u_{i,t-1}) - 2 \text{Cov}(u_{it}, u_{i,t-1}) \\ &= 2\sigma_u^2.\end{aligned}$$

For adjacent differenced errors,

$$\begin{aligned}\text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) &= \text{Cov}(u_{it} - u_{i,t-1}, u_{i,t-1} - u_{i,t-2}) \\ &= -\text{Var}(u_{i,t-1}) = -\sigma_u^2.\end{aligned}$$

So differencing mechanically creates negative serial correlation even when the original disturbances were serially independent.

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Appendix: Double demeaning removes unit and time effects

Start from

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it}.$$

Take the unit mean, time mean, and grand mean:

$$\bar{y}_i = \bar{x}'_i\beta + \alpha_i + \bar{\lambda} + \bar{u}_i,$$

$$\bar{y}_t = \bar{x}'_t\beta + \bar{\alpha} + \lambda_t + \bar{u}_t,$$

$$\bar{y} = \bar{x}'\beta + \bar{\alpha} + \bar{\lambda} + \bar{u}.$$

Now compute

$$y_{it} - \bar{y}_i - \bar{y}_t + \bar{y} = (x_{it} - \bar{x}_i - \bar{x}_t + \bar{x})'\beta + (u_{it} - \bar{u}_i - \bar{u}_t + \bar{u}),$$

because the α_i and λ_t terms cancel exactly.

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Appendix: Demeaned errors are mechanically correlated

Let

$$\ddot{u}_{it} = u_{it} - \bar{u}_i, \quad \bar{u}_i = \frac{1}{T} \sum_{t=1}^T u_{it},$$

with serially uncorrelated u_{it} and variance σ_u^2 . Then

$$\text{Var}(\bar{u}_i) = \frac{\sigma_u^2}{T}, \quad \text{Cov}(u_{it}, \bar{u}_i) = \frac{\sigma_u^2}{T}.$$

Hence

$$\text{Var}(\ddot{u}_{it}) = \text{Var}(u_{it}) + \text{Var}(\bar{u}_i) - 2 \text{Cov}(u_{it}, \bar{u}_i) = \sigma_u^2 \left(1 - \frac{1}{T}\right).$$

For $t \neq s$,

$$\text{Cov}(\ddot{u}_{it}, \ddot{u}_{is}) = 0 - \frac{\sigma_u^2}{T} - \frac{\sigma_u^2}{T} + \frac{\sigma_u^2}{T} = -\frac{\sigma_u^2}{T}.$$